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How Chatbots Augment Human Intelligence in Customer Services: A Mixed-Methods Study

Xiaolin Lin[a], Xuequn Wang[b], Bin Shao[c], and Joseph Taylor[a]

[a]Department of Information Systems and Business Analytics, College of Business, California State University, Sacramento, Sacramento, CA, USA; [b]School of Business and Law, Edith Cowan University, WA, Australia; [c]Paul and Virginia Engler College of Business, West Texas A&M University, Canyon, TX, USA

ABSTRACT

Although artificial intelligence (AI)-enabled chatbots have been increasingly incorporated into firms' business processes, such as customer service, they are still ineffective at handling unstructured tasks. Therefore, organizations might adopt chatbots to cut down the workforce for structured tasks and augment the remaining employees in unstructured tasks. Although there are substantial workforce implications for rebalancing work activities, few studies have assessed chatbots from the augmentation angle by focusing on employees' performance, perhaps because of the lack of a theoretical foundation and research on this new phenomenon within organizations. Using a mixed-methods design, we developed a model illustrating the impact of chatbots on perceived work performance from the information technology artifact perspective. We then used the qualitative study to conceptualize contextualized technology affordances to enhance customer service, support decision-making, and improve business agility and chatbot support (i.e. informational and emotional support) in the context of chatbots. We then used the quantitative study to validate context-specific variables and the research model. The results showed that three types of chatbot affordances are positively related to post-adoptive chatbot use, through which chatbots provide employees with informational and emotional support and improve perceived work performance. Our study makes an original contribution to the development of a theory about new IT phenomena related to the impact of chatbots on work performance. Our results also provide suggestions for organizations regarding how to integrate chatbots to improve work performance.

KEYWORDS

Artificial intelligence; AI; chatbots; chatbot affordances; chatbot use; chatbot support; work performance; human-AI augmentation; customer service; unstructured tasks

Introduction

Consider the following customer service dialogue:[1]

> Hey James, it's Alexander from Tesla Motors. How's your day going so far?

CONTACT Xuequn Wang xuequnwang1600@gmail.com School of Business and Law, Edith Cowan University, 270 Joondalup Dr., Joondalup, WA 6027, Australia

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> It's going pretty good, my man. How about yourself? You said from Telsa, right? The car company?
>
> Yeah, that's right. I'm calling from Tesla Motors, the car company. Did you have any trouble or questions while customizing your car on our website?
>
> Uh no. We're just kinda shopping around and, uh you know, just taking a look.
>
> Gotcha. That makes sense. Sometimes we're just exploring our options before making decisions, and I totally respect that . . .

Although the dialogue sounds normal, it is actually an example of how chatbots, developed by Air AI, can provide customer services over the phone, when customers might not know they are talking to chatbots. Chatbots, as powerful artificial intelligence (AI) tools,[2] have been widely used by organizations as innovative ways to support marketing processes and enhance customer service [23, 38, 48, 63]. The value of e-commerce transactions supported by chatbots is predicted to reach $112 billion by 2023 [24]. Chatbots can automatically respond to customers' standard questions, even during nonbusiness hours [44]. A recent report shows more than 80% of internet consumers have interacted with chatbots at some point [10].

Despite recent promising progress, the current AI-enabled chatbots are still ineffective in dealing with unscripted and unstructured tasks [8, 36]. Unstructured tasks refer to those tasks dealing with incomplete or uncertain information, or involving tacit knowledge [12]. For example, Jia et al. [36] assessed the unstructured task of sales persuasion, which deals with incomplete customer information and requires tacit knowledge of persuasion. The report from Bleu [10] showed chatbots still cannot handle at least 30% of conversations with consumers. Another report showed the chatbot, Tessa, deployed to (unintentionally) replace helpline employees, provided harmful advice regarding eating disorders [5]. Indeed, about 75% of consumers believe chatbots cannot deal with complicated requests, and 30% of consumers might abandon their purchases after negative experiences with chatbots [57]. The recent literature has also found chatbots are less effective than top-skilled employees in sales persuasion [36]. Therefore, completely replacing employees with chatbots in customer services is risky and may lead to negative impacts, such as reduced levels of services.

To avoid such negative impacts, organizations should augment, rather than completely replace, employees. Firms might consider cutting down the workforce for scripted and structured tasks (e.g., introducing products and probing consumers' interests [36]) and use chatbots to help augment the remaining employees. Chatbots can collect customers' data during their interactions and generate important insights [69]. As a result, they might help employees acquire knowledge about their customers' profiles and buying patterns and could target consumers with the correct products or services at the right time [18, 75, 80].

Another recent survey found that 79% of customer service employees believe that chatbots can improve their customer service skills [5]. However, our review (Online Supplemental Appendix 1) showed that few studies have assessed chatbots from the augmentation angle by focusing on employees' performance. Therefore, to fill the research gap with important theoretical and practical implications, our study focuses on the following questions: *How do chatbots augment customer service employees, and what are the effects of such augmentation on job performance?*

To ground our study, we drew on the information technology (IT) artifact, defined as, "the integration of the processing logic found in computers with the massive stores of databases and the connectivity of communication networks" [3:394]. Markus and Silver [51] proposed an approach for characterizing the IT artifact via three components: technical objects, functional affordances, and symbolic expressions.

The technical object component pertains to the IT artifacts themselves. Functional affordances are defined as, "the possibilities for goal-oriented action afforded to specified user groups by technical objects" [51:622] and can be understood as necessary conditions for IT usage. Symbolic expressions are defined as, "the communicative possibilities of a technical object for a specified user group" [51:623] and can be understood as necessary conditions for user interpretations of IT artifacts and the consequences resulting from those interpretations. Therefore, we argue that the impacts of IT can be established from both the technical and the social components (i.e., user interpretations), which can be further captured by chatbot affordances and support.

Chatbot support captures symbolic expressions in that it deals with chatbots' relationships with employees and reflects chatbots' abilities to convey messages so they can support employees' tasks. In this regard, chatbots can be viewed as a new IT artifact providing functional affordances that motivate users to adopt chatbots (IT use) and generate the impacts of chatbot usage (IT impact [7]). In addition, neither of the functional affordances of chatbots, nor chatbot support alone, can explain the effects of chatbot usage because no single component of the IT artifact can work for every technology [51]. Together with other influences, the IT artifact can explain the behavioral effects of chatbot use. Thus, combining the two aspects (i.e., technology affordances and chatbot support) can provide novel insights into the use of chatbots and their impacts on work performance.

To answer our research question, we adopted a mixed-methods approach following the guidance provided by Venkatesh et al. [76, 77]. This approach aims to develop a substantive theoretical framework for our study and to achieve scientific rigor and realism [4].

Our work makes several key contributions. First, the current literature concerning chatbots from the employee perspective has mainly focused on employees' attitudes and use of chatbots [44, 79]. As a result, empirical evidence regarding how chatbots augment employee work performance is lacking. Because the current forms of AI-enabled chatbots are still ineffective in dealing with unscripted and unstructured tasks [8, 36], it is of both theoretical and practical importance to provide an enhanced understanding of the impacts of chatbots on work performance from the augmentation angle and explain the mechanism through which chatbots can augment employees and improve their work performance. Our study can contribute to the ongoing debate regarding whether AI should replace, or augment, our workforce [20, 60]. Our post-hoc analysis (Online Supplemental Appendix 12) reveals that chatbot affordances have differential effects on chatbot use, indicating various pathways to augment employees.

Second, the focus on chatbots provides the opportunity to develop a contextualized understanding [32] of work performance in the field of IS, especially in light of technology affordances and social support from the IT artifact perspective. Technology affordances have been recently examined in the information systems (IS) field (e.g., [15, 39]). It is of theoretical importance to conceptualize technology affordances in the context of chatbots to advance and provide

a contextualized understanding of the technology affordance theory [32]. This conceptualization helps illustrate how specific types of chatbot affordances augment employees and impact the employees' job performance, thus contributing to the literature.

Further, the unique characteristics of chatbots allow us to assess chatbot support, advancing the traditional IT artifact to include informational and emotional support. Unlike the literature focusing on social support between individuals (e.g., [42, 82]), our study demonstrates how chatbots can also facilitate social support, thus generating significant impacts on employees' work performance. Our study depicts how AI has changed the way employees acquire social support for employee performance.

Literature Review and Theoretical Background

Chatbots: A Review from a Technological View

According to the literature, there are two types of chatbots based on the algorithms adopted: rule-based and AI-enabled (e.g., [2, 14]).[3] Rule-based chatbots generate responses from predefined rules [58]. Accordingly, rule-based chatbots operate by matching individuals' inputs to rules and selecting predefined answers with pattern-matching algorithms, and they do not generate new information or knowledge [58]. In contrast, AI-enabled chatbots are empowered by advanced AI techniques, such as deep learning, neural networks, and natural language processing (NLP) [19, 36], which have been found to be effective means of processing and synthesizing the data required for customer support. To train themselves, they require significant amounts of data, potentially up to terabytes, and they can learn from their interactions with individuals. Thus, AI-enabled chatbots can enable human-like communication by considering conversational contexts, learning from previous conversations, and generating natural and appropriate responses [2]. Examples of AI-enabled chatbots include Amazon Lex, Messenger Platform, and IBM watsonx Assistant.[4]

The aforementioned discussions suggest rule-based chatbots only provide predefined answers and cannot generate new knowledge to augment employee performance [58]. In contrast, AI-enabled chatbots can communicate with consumers in a natural and a human-like way by considering the conversational context. They can also continue integrating new sources of data, evolving by learning from previous interactions with consumers and generating new insights [27]. Such new insights can help marketing employees gain expansive knowledge of current customer demands, product challenges, and marketing changes, allowing employees to effectively perform customer service activities [80]. Therefore, AI-enabled chatbots can effectively support employees' tasks and augment human intelligence, and they are the focus of our work.

Our study selects customer services as the research context, in which completely replacing employees with chatbots can be risky and lead to negative impacts. We focus on AI-enabled chatbots that have been widely adopted in various industries for use as both customer-facing and internal tools. One example is Amazon Lex.[5] Chatbots enabled by Amazon Lex can be informational or transactional, can connect with websites—social media, or other applications—and can follow generative interaction styles. Such a focus is also consistent with the recent literature on chatbots (e.g., [36, 48]).

Related Studies About Chatbots

The literature has investigated how chatbots can improve customer service from the perspectives of both the consumer and the employee, as shown in Online Supplemental Appendix 1 (see Table 1.1). From the consumer perspective, the literature has attempted to investigate consumers' perceptions of chatbot use and engagement by exploring chatbot features. Blut et al. [11] conducted a meta-analysis to clarify the underlying mechanisms through which anthropomorphism influences customer use intention. Anthropomorphism is vital to influence customers' perceptions and behaviors [65]. Saffarizadeh et al. [61] reported reciprocal self-disclosure increases anthropomorphism, which in turn increases cognition-based and affect-based trustworthiness. Chatbots can improve their interactions with customers by employing effective messages. For example, Schuetzler et al. [64] found that individuals perceive skillful chatbots to be more socially present and anthropomorphic. More recently, Chandra et al. [16] revealed that human-like interactional competencies can facilitate user engagement via trust. Han et al. [30] showed that AI-expressed positive emotions influence service evaluations via emotional contagion and expectation–disconfirmation. Other studies indicate that chatbots can better understand consumers and improve customer service because of the technology's unique features. Shumanov and Johnson [69] showed that a consumer's personality can be predicted based on their interactions with chatbots, and engagement can be enhanced by matching the consumer's personality with the chatbot's personality (inferred by their response language).

Considering the employee perspective, the literature focuses on the use of chatbots to provide customer service and examines their effectiveness. For example, Behera et al. [6] showed that chatbots' information, system, and service quality improve the customer experience, which in turn increases employees' intentions to use chatbots for customer service. Lin et al. [44] reported that chatbot affordances for personalization and automatability can increase chatbots' effectiveness in interacting with customers. Wang et al. [80] demonstrated that the use of chatbots can improve customer service by creating agility. These studies suggest the literature has provided some solid understanding of the impacts of chatbot use on customer service. Nevertheless, there is still a need to advance the literature and enrich our understanding of the implications of chatbots and their impacts on business. From the IT artifact perspective, we aim to investigate how chatbots can augment employees' work performance.

Work Performance

The IS literature has contextualized work performance in various ways. At the organizational level, aggregate investment has long been identified as a mechanism for boosting productivity [31, 37, 72]. However, substantial variation has been observed in the benefits of technology for organizations [22].

We focused on work performance at the individual level, examining how IS might contribute to employees' perceptions of their own performance. Technology can make a difference at the individual level, with the potential for positive results. To increase the *quantity* of work completed, IS can enable data sharing and real-time feedback, potentially improving work performance through the gamification of tasks [71]. To increase the *quality* of work completed, the standardized capabilities and contextualized communication

provided by IS can enhance communication across cultural groups [59]. Our study further examined the influence of technology on individual work performance. We focused on how perceptions of individual work performance change through the greater inclusion of the emerging technical capabilities that chatbots facilitate.

Central to our understanding of individual work performance was the framing of our dependent variable. In the context of knowledge work, Shirish et al. [67] identified two aspects of work performance: efficiency and creativity. Accordingly, we included both productivity and creativity in our conceptualization of employee work performance [83]. Within the context of our study, work efficiency reflected employees' perceptions of the quality and quantity of work completed. In contrast, work creativity was demonstrated by continuous improvement and experimentation to improve work processes [67]. When processes are continually enhanced, overall work performance is positioned to improve over time.

IT Artifact: Technology Affordance and Social Support

Drawing on the concept of the IT artifact in the literature, we have identified technology affordance and social support as the primary components of the artifact of chatbots to clarify their usage and their impacts. As explained below, we focused on higher-level affordances to better link chatbots with business practices [25].

Technology Affordance

Technology affordance is defined as, "the mutuality of actor intentions and technology capabilities that provide the potential for a particular action" [50, p. 39]. Because actors may perceive the same technology differently, a technology affordance emerges when actors realize that technological features can be used in certain ways to support their goals [28, 40]. The literature has conceptualized technology affordance in various contexts (see Online Supplemental Appendix 1, Table 1.2).

Fayard and Weeks [25] suggested that affordance needs to focus more on practices enabled through technology. Bygstad et al. [13] also argued that affordance (focusing on technology features) should not be the endpoint, and researchers need to abstract affordance into more complex mechanisms. The literature has proposed first-order versus second-order affordances [39]. The argument is that second-order affordances (e.g., affordances focusing on business practices) function at higher levels than first-order affordances (e.g., affordances focusing on technology features [13], and second-order affordances can be actualized only after first-order affordances are actualized (i.e., when certain technology features are used by individuals; [70]. For example, interacting with peers can function as the first-order affordance of enterprise social media. Interacting with peers can then actualize a second-order affordance, such as building relationships with peers [39].

The above discussion suggests two primary differences between the first- and second-order affordances of IT: first-order affordances capture the unique features of IT, which can be used in certain ways; moreover, first-order affordances can be readily actualized when people use IT because these first-order affordances largely depend on the design features.

Second-order affordances are typically actualized when people use the relevant IT features (i.e., first-order affordances) to accomplish their goals, and they are

operationalized by the corresponding practices or processes. For example, Lin and Kishore [43] proposed the social media affordance for social relationships and suggested that it can be actualized when people use relevant social media features (e.g., comments) to engage in developing and maintaining relationships with others. As a result, second-order affordances can better capture practices during the use of IT. Because second-order affordances might not be readily actualized, and their actualizations depend on the actualization of other (first-order) affordances, a qualitative study is warranted to conceptualize key chatbot affordances via in-depth interviews with employees (Strong et al., 2014).

From the organizational perspective, second-order affordances can help us better understand how IT helps organizations achieve certain business practices. De Luca et al. [21] proposed three types of second-order affordances in the context of big data marking: customer behavior pattern spotting, real-time market responsiveness, and data-driven market ambidexterity. Despite important progress in the research, affordance conceptualization in AI in general and in the context of chatbots in particular is still limited. One exception is Lin et al. [44], who proposed three types of chatbot affordances: availability, personalization, and automatability. However, their study dealt with first-order rather than second-order affordances. Although the conceptualization from Lin et al. [44] provided important insights into chatbots' features, the link between those features and business practices is less clear. Therefore, further studies are needed to assess the second-order affordances of chatbots to better clarify the role of chatbots in business processes. Our study fills the gap by conceptualizing the second-order affordances of chatbots and clarifying how they affect employees' perceived work performance. Because only a few studies deal with chatbot affordances, especially at the higher order, we followed the mixed-methods approach and conceptualized the second-order affordances of chatbots based on qualitative interviews.

Social Support

Our study conceptualizes chatbot support using the concept of social support. Social support can be defined as the exchange of informational and emotional messages via verbal and nonverbal contact between individuals [43, 68]. IS researchers have assessed social support's formation and impacts on emerging technologies. Lin et al. [47] delineated social support in social media by identifying its two forms of informational and emotional support. In the context of social commerce, social support has been used to investigate the impacts of social media on consumer behavior, such as reviews, comments, referrals, recommendations, and purchases (e.g., [29, 42, 46, 66]). In the context of health care, social support has also been demonstrated as an important factor that influences patients' self-care and outcomes [34, 43, 82].

These studies confirmed the important role of social support in understanding the impact of IT. Furthermore, they suggested social support should be divided into different types, such as informational and emotional support, which we argue that chatbots can also facilitate. Next, we focused on understanding the nature of chatbot support in the use of chatbots within organizations through phase one of the mixed-methods design.

Mixed-Methods Methodology

Prior studies have suggested a mixed-methods approach can achieve scientific rigor and realism through both qualitative and quantitate research designs [73, 76]. There are two reasons we chose the mixed-methods approach. First, our research context (i.e., the use of AI-enabled chatbots to promote work performance) is difficult for existing theories to delineate and explain [4, 76]. Our research question attempts to extend existing theories about "chatbot affordances" and "social support" by incorporating the unique contextual features of chatbots [32]. These two topics are relatively underexplored in the chatbot context. A mixed-methods approach is needed to develop a substantive theoretical framework [77].

Second, we can improve our study's scientific rigor and realism with a mixed-methods design [4] by conducting both qualitative and quantitative research. It is appropriate to answer both confirmatory and explanatory research questions [77, 81]. Thus, the mixed-methods approach can provide a global view that helps us achieve a robust understanding of the relatively new phenomenon in our context. Our research questions and contexts suggest the mixed-methods approach is appropriate [76, 77].

The use of the mixed-methods design can be understood as the developmental approach [33, 73, 76]. The qualitative study can be used to articulate the theoretical foundation and develop the key concepts to explain the impacts of chatbots within organizations. The quantitative study can be used to test the research model generated by the prior qualitative study. Accordingly, we conducted the qualitative study during the first phase, followed by the quantitative study in the second phase. The developmental approach was necessary for three reasons. First, research into the impacts of chatbots on their organizations is limited. Second, the concept of social support is relatively new in the IS field, and the emphasis has usually been directed toward online shopping [42, 46] and health care [34, 82]. Third, few studies have conceptualized second-order chatbot affordances or examined the role of chatbots in business practices from the employee perspective.

From an epistemological perspective, we used the interpretive perspective in the qualitative study and then focused on a positivist perspective in the quantitative study. Based on the guidelines provided by [76, 77], we first found general theories of affordances and social support. We then conducted a qualitative study to contextualize our research model and articulated constructs of chatbot affordances and social support by carefully analyzing the qualitative data. Third, we tested and validated our research model through a quantitative study based on the survey data. Finally, we incorporated the results of our qualitative and quantitative studies to contribute to an overall understanding of the mechanism through which chatbots impact employee performance. Online Supplemental Appendix 2 elaborates on our decision choices about the mixed-methods design approach. Online Supplemental Appendix 3 shows how we established the mixed-methods design.

Phase 1: Qualitative Study

Data Collection and Data Analysis

We collected our data from in-depth interviews with employees. We selected full-time employees who had experience using AI-enabled chatbots in the workplace and gave them the freedom to express their views. We drew the sample from MBA students in a Western

public university and sent an email invitation; those who were full-time employees currently using AI-enabled chatbots to support customer services were selected. Upon successful completion of the interviews, they were awarded 1% of the total grade in their course. We successfully recruited 22 participants. We ensured design validity by including participants from a variety of industries and of different ages and occupations. We conducted semi-structured interviews on Zoom that ranged from 25 to 40 minutes. We recorded all interviews with permission. We summarized their profiles and their respective organizations in Online Supplemental Appendix 4 (Table 4.1).

The theoretical framework of the IT artifact (more specifically, social support and IT affordances) provided guidance for our data analysis with the aim of advancing the theoretical concepts in the chatbot context. This helped us find the emergent themes and generate new insights [62] into the core concepts of chatbots in improving work performance. Specifically, we identified the relevant concepts through systematic reading and rereading of the interviews and labeling and relabeling of the concepts [78]. Following a multiple classification schedule (e.g., [9]), we first independently coded each response into one or more categories. In follow-up discussions, we arrived at a consensus to confirm the codes and findings. The overall process was iterative and roughly resembled a constant comparative analysis to reach saturation. We summarized the emergent themes and supporting empirical data with our initial codes and consensus in Online Supplemental Appendix 4 (Table 4.2).

To ensure the merit of the qualitative inquiry, we followed the guidelines provided by Venkatesh et al. [76] and took measures, including design adequacy and analytic adequacy. In addition, we ensured inferential validity by validating the identified variables from the qualitative study in the quantitative survey. Details of these measures can be found in Online Supplemental Appendix 3.

Research Findings

In this section, we explain our key findings. Online Supplemental Appendix 4 (Table 4.2) contains a more detailed report. We first demonstrate that the use of chatbots is an increasingly important IT phenomenon for improving employee job

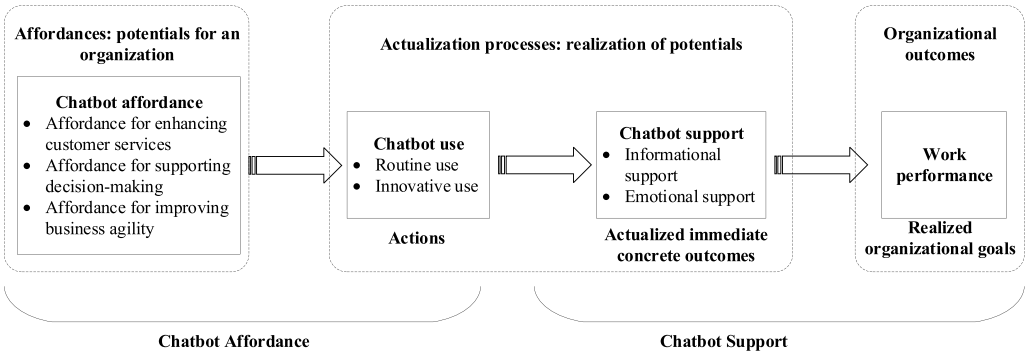

Figure 1. The conceptual framework: An information technology (IT) artifact perspective. Source: Strong et al. 2014.

performance. We then identify the core constructs that illustrate the mechanism of how chatbots can affect job performance based on our analysis of qualitative data. Our conceptual framework, developed from the qualitative study, is shown in Figure 1, and is built on the framework of Strong et al. [70]. Our results represent how chatbots can be used [50]. Such affordances then lead to actualization processes [70]. In the context of customer service, employees can engage in various actions (i.e., chatbot use) to achieve immediate concrete outcomes (i.e., chatbot support). Such outcomes ultimately generate important organizational outcomes and help realize organizational goals (i.e., enhanced work performance [70]).

New IT Phenomenon: Chatbot Drives Employee Performance

The interviewees expressed their opinions about the important role of chatbots in driving their performance. In today's rapidly changing environment, emerging technologies have been increasingly incorporated into business processes and can thus bring benefits to both employees and organizations. Therefore, employee performance can be improved by incorporating chatbots within organizations. For example, one interviewee commented,

> When I do have to rely on [a] chatbot, first of all, it's very efficient. It helps me save time. I am more productive. Secondly, it gives me higher accessibility. I can troubleshoot the client request whenever I need to. It helps me manage my own time.

In this way, chatbots can effectively enhance employee work performance by saving time in resolving problems, thus highlighting the importance of the new phenomenon in business practices and motivating us to further investigate the mechanism through which chatbots improve employee performance. The process can be explained by the different types of chatbot affordance and support we found in the analysis of the qualitative data.

Chatbot Affordances

The literature has shown IT affordance plays an important role in the process of IT adoption. Our qualitative study identified three main second-order chatbot affordances. Second-order affordances can help us better understand how chatbots help organizations achieve certain business practices via their unique features. Our interview data demonstrated that technological features lead to different types of chatbot affordances. One interviewee stated, "The bot helps the company to address the customers' needs right away and improve customers' satisfaction." This indicates that chatbots can empower companies to enhance their customer service to improve customer satisfaction, a result we identify as the *affordance for enhancing customer service*.

Furthermore, our interview data revealed an *affordance for decision-making*. For example, one interviewee stated, "Chatbots help the triage process and direct applicants to the right place faster." Chatbots can help employees make decisions about the process and improve service quality. Chatbots can also provide insights into each individual customer by taking into account unique circumstances, allowing employees to make informed decisions that are beneficial to the customers. Another interviewee stated, "In some cases, it really reduces the time of communication because you don't need to wait around for others to tell you who to contact. I can find the right person by using [a] chatbot. I don't need to search around and call many people."

In addition, we observed an *affordance for improving business agility* by analyzing the interview data. Business agility refers to the degree to which companies can utilize their resources (e.g., technology) to efficiently identify opportunities and alleviate threats [52]. For example, one interviewee said, "Nora allows us to provide a better product to our patients," indicating that chatbots can convey information about customers, which helps organizations learn about customers' needs, thus assisting in customer retention and improving business agility. This insight also suggests chatbots offer the company the opportunity to respond to the market by improving the quality of their products. Another interviewee relayed, "Additionally, we use them to collect and analyze consumer data, allowing the business to gain insights into customer preferences and behavior, which informs marketing strategies and product development." Organizations can improve their marketing strategies with the help of chatbots to enhance business agility.

Accordingly, our qualitative interviews identified three types of high-level chatbot affordances: enhancing customer service, supporting decision-making, and improving business agility. This list is consistent with a systematic review from Achmat and Brown [1], who found that AI affordances, such as customer end-user interaction, proactively anticipate and react to changes and assist in decision-making. In our study, the affordance for enhancing customer service refers to the degree to which employees perceive that chatbots provide the possibility to improve interactions with customers. The affordance for supporting decision-making is defined as the degree to which employees perceive that chatbots provide the possibility to generate useful insights and facilitate good decisions. The affordance for improving business agility refers to the degree to which employees perceive that chatbots provide the possibility to respond quickly to market changes. Essentially, because one of the main tasks of marketing employees is to provide customer service, the affordance for enhancing customer service indeed reflects the augmentation of employees. To further understand these high-level chatbot affordances, we provide more results of our qualitative data in Online Supplemental Appendix 5, showing how each affordance is related to relevant technological features.

These chatbot affordances facilitate the process of chatbot adoption. For example, one interviewee stated, "Most of the time, [a] chatbot is used to create requests, so the chatbot is part of self-service to [do] that. It optimizes the process and minimizes the calls that reach the help desk." Chatbots can help improve customer service, can be integrated into the process of customer service, and can be used routinely to create customer requests.

As firms become more familiar with chatbot affordances, they also engage in innovative uses. One interviewee said, "We have created [a] weather service channel for users, also issue resolution if customers have issues, which is the incident management process." In this case the firm created new chatbot-enabled services (i.e., a weather service channel and issue resolution). These new services are also based on chatbot affordances, such as enhancing customer service. To summarize, our qualitative study identified three types of second-order chatbot affordance that facilitate chatbot adoption and use.

Chatbot Support

The qualitative data also revealed that the use of chatbots can provide both informational and social support for employees. For example, one interviewee said,

> If I, personally speaking, am not the best person to remember things, I am able to get resources and documentation from the chatbot. I am able to bookmark resources and documents from [the] chatbot and use them for the same troubleshooting. So it's continuous support. Also, in that process, I might not have to go back to the documentation because it improves my own knowledge to become more technical.

The interviewee received useful information and knowledge for resolving problems and assisting with job tasks, which can be understood as informational support. Another interviewee also reported the informational support they receive from the use of chatbots, "For [the] service side, I would think [of a] chatbot as a backup knowledge base."

In addition, one interviewee revealed that chatbots can provide employees with emotional support by building employees' confidence to accomplish their tasks, as well as by providing access to more resources: "I feel more confident to do my job. For example, it allows more effective communication during [a] virtual meeting. It also reduces the stress level of me doing troubleshooting during [a] virtual meeting." In addition, the use of chatbots can also make employees feel comfortable. Another interviewee commented, "I feel more comfortable doing my job because I can do my jobs faster and easier."

In summary, the aforementioned research results align well with social support theory, which indicates chatbot support can be understood as two types of support, informational and emotional. Therefore, it is important to delineate and contextualize the concepts of social support in chatbots from the employee perspective by differentiating between informational and emotional support.

Furthermore, our interviewees highlighted the role of chatbot support in influencing their work performance. One interviewee stated, "The information or knowledge that can be conveyed through use of the chatbot can help with completing work more quickly or with more knowledge of the specific situation." Another interviewee said, "I feel more confident to do my job because it helps me see the issue right away."

Chatbot support can lead to enhanced work performance with its technical capability to augment human performance. This technical capability to function without human intervention marks a distinction from previous research, which had focused on demonstrating the efficacy of human-generated social media content to provide social support.

Research Model and Hypotheses Development

Based on the research findings of the qualitative study, we developed our research model by identifying key chatbot affordances and by conceptualizing and operationalizing informational and emotional support. We explain the constructs in detail and then present our research model in the following sections.

Chatbot Affordances

As shown in Online Supplemental Appendix 1 (see Table 1.2), chatbot affordances have not been well conceptualized or measured. To the best of our knowledge, only one study has attempted to clearly conceptualize lower-level chatbot affordances by capturing the technical features [44]. However, few studies have conceptualized second-order affordances in the context of chatbots. Our study aims to fill this gap by operationalizing the second-order

Table 1. Second-order chatbot affordances (our study) actualized by first-order chatbot affordances.[a]

Second-Order Affordance	Definition	Relevant First-Order Affordance
Affordance for enhancing customer services	The degree to which chatbots provide the possibility to improve interactions with customers	• Automatability • Personalization • Availability
Affordance for supporting decision-making	The degree to which chatbots provide the possibility to generate useful insights and make good decisions	Personalization
Affordance for improving business agility	The degree to which chatbots provide the possibility to respond quickly to market changes	Personalization

Note: [a]Lin et al. 2022 [44].

affordances of chatbots, which do a better job of linking chatbots with business practices [25].

Accordingly, our study conceptualizes second-order chatbot affordances based on the analysis of the qualitative interviews (see Table 1). We also linked the second-order affordances to the first-order affordances (i.e., automatability, personalization, and availability) from Lin et al. [44], to demonstrate how our study is related to the literature. Online Supplemental Appendix 5 also provides complementary results to show how our identified high-level affordances can be facilitated by technological features. When chatbots can automatically provide personalized responses to customers at all hours, they can enhance customer service. Personalized answers from chatbots can also help improve business agility because they can help employees better understand customers' personalized needs and preferences. Lin et al. [44:47] defined automatability as, "the extent to which marketing employees believe that chatbots offer the opportunity to respond to customers' questions automatically." Therefore, we did not link automatability to the support of decision-making or improvement in business agility because it depends on whether chatbots can automatically respond to customers rather than whether chatbots can automatically generate new knowledge. Such linkages further demonstrate that our developed chatbot affordances are appropriate and reflect how employees use chatbots to actualize higher-level affordances via lower-level ones [70]. Our study can bridge the literature dealing with first-order chatbot affordances in business practices and provide useful insights regarding second-order affordances.

Routine and Innovative Use

Technology can be used in different approaches, and research has found that employees can engage in both routine and innovative use of chatbots. Here, routine use refers to, "employees using IS in a routine and standardized manner to support their work" [41:659]. Routine use occurs when employees integrate technology, such as chatbots, into their normal work activities [41]. Marketing employees can use chatbots to provide customer service and answer consumers' standard questions [80]. In our qualitative study, our interviewees stated that chatbots had become an essential part of customer service.

In contrast, innovative use deals with, "employees' discovering new ways to use IS to support their work" [41:659]. In innovative use, employees utilize new features of the technology to develop creative approaches to further realize the value of the technology [35]. In our qualitative study, our interviewees stated they use chatbots to provide other services such as issue resolution.

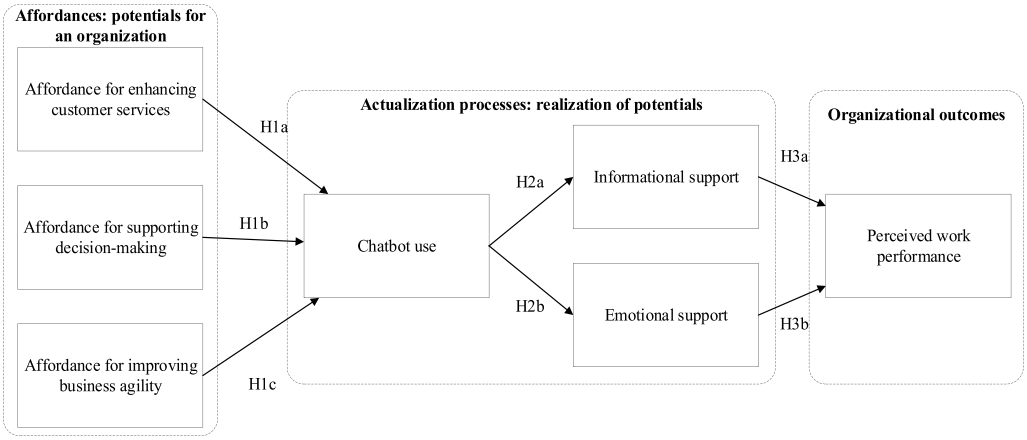

Figure 2. Research model.

Informational Support and Emotional Support

Informational support refers to the provision of useful information and knowledge that can help others solve problems and accomplish their tasks [42, 46]. Pérez et al. [55] demonstrated that chatbots can generate new knowledge from data analysis based on AI algorithms. Chatbots are trained based on previous conversations with customers. New insights and knowledge can be generated to help understand customers and their needs, acquire new customers, and optimize pricing via data analysis powered by AI algorithms [54].

Emotional support refers to verbal and nonverbal ways of showing care, happiness, and understanding [46, 82]. In the context of our study, chatbots can build employees' confidence to perform tasks and solve problems because chatbots have access to useful information. By solving problems more efficiently, employees can reduce their stress while providing customer service, thus receiving emotional support.

Research Model

Figure 2 presents our research model. We employed the IT artifact as the mechanism through which chatbots can improve employee performance within organizations. Also note that our research model is fully based on the framework developed by Strong et al. [70]. Specifically, our model argues that chatbot affordances lead to chatbot use (i.e., action in actualization processes), which then results in chatbot support (i.e., actualized immediate concrete outcomes in actualization processes). Then, chatbot support ultimately enhances employees' perceived work performance (i.e., helps realize organizational goals). Online Supplemental Appendix 6 presents detailed arguments for each hypothesis, based on our qualitative results and the relevant literature.

Although Strong et al. [70] also asserted that actualization processes can have feedback effects on affordances, our study focuses on how affordances influence action in actualization. Such a focus is consistent with the main relationship between an affordance and its actualization. As argued by Strong et al. [70], identifying an affordance is the initial step toward understanding how chatbots impact organizations whereas theorizing the

actualization process can provide deeper insights into organizational changes raised by chatbots. Therefore, our theory about the effects of chatbot affordances on chatbot use (rather than vice versa) can help researchers better understand how chatbots generate organizational changes.

Phase 2: Quantitative Study

Data Collection Procedure and Sample

We hired a survey company to recruit our participants, focusing on American employees. The survey company had access to national panels that included people from different backgrounds. The company sent an email including the study description and survey link to the participants via systematic sampling. Participants qualified for our study if they were marketing employees and provided customer service with AI-enabled (rather than rule-based) chatbots. In total, we received 203 valid responses (Online Supplemental Appendix 7). On average, the participants had worked for their current companies for 7.50 years (standard deviation: 3.73).

Measures

We adapted our measures from the literature wherever possible (Online Supplemental Appendix 9). Specifically, we adapted the items for informational and emotional support from Lin et al. [46]. We measured perceived employee performance as a second-order formative construct, including perceived workplace productivity and perceived workplace creativity, items we adapted from [67]. We measured chatbot use as a second-order formative construct, including routine and innovative use, items we adapted from [80]. Regarding items of chatbot affordances, we followed standard procedures of item development [49, 74]. The entire process is described in Online Supplemental Appendix 8. We measured all items with a 7-point Likert scale from 1 (strongly disagree) to 7 (strongly agree).

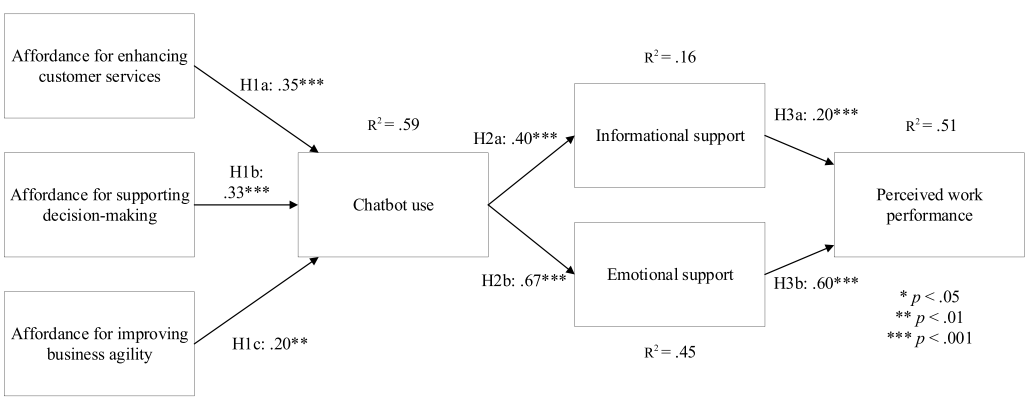

Figure 3. Research model with regression analysis.

Data Analysis and Results

We first assessed common method bias following Podsakoff et al. [56], but we did not find it a serious issue (Online Supplemental Appendix 10). Our next step involved assessing our measures with confirmatory factor analysis (Online Supplemental Appendix 11). Overall, our measures had good construct validity. Next, we tested the hypotheses with regression analyses (Figure 3). The affordance for enhancing customer service ($\beta = .35$, $p < .001$, $f^2 = .21$), affordance for supporting decision-making ($\beta = .33$, $p < .001$, $f^2 = .12$), and affordance for improving business agility ($\beta = .20$, $p < .01$, $f^2 = .05$) were positively related to chatbot use, supporting H1a, H1b, and H1c. Chatbot use was positively associated with informational support ($\beta = .40$, $p < .001$, $f^2 = .19$) and emotional support ($\beta = .67$, $p < .001$, $f^2 = .82$), supporting H2a and H2b. Last, both informational ($\beta = .20$, $p < .001$, $f^2 = .12$) and emotional support ($\beta = .60$, $p < .001$, $f^2 = .62$) had positive effects on perceived work performance, supporting H3a and H3b. Relationships had at least a small effect size, indicating these significant effects were not trivial. We then tested the mediation effects depicted by the model, and the results (Online Supplemental Appendix 12) provided further support for our model.

To gain a more nuanced understanding of the relationships in our model, we also conducted robustness checks (Online Supplemental Appendix 12). First, we assessed the antecedents and consequences of routine versus innovative use. Second, we examined how informational and emotional support influence workplace productivity versus workplace creativity. Third, we reran our analysis with structural equation modeling. Lastly, we controlled for various variables.

Discussion

Using our mixed-methods approach, we aimed to understand the mechanism through which chatbots can improve customer service employees' perceived work performance from the IT artifact perspective. The research results of the quantitative study indicated that social support in the context of chatbots can be divided into two types: informational and emotional. Our qualitative study suggested interviewees received different kinds of support through the use of chatbots within organizations, indicating the possibility of multiple dimensions of chatbot support. The interviewees' quotations revealed both informational and emotional support. The quantitative study further corroborated these kinds of chatbot support by confirming the discriminant validity between informational and emotional support. Those findings suggested that informational and emotional support are two conceptually distinct variables that drive employee performance.

In addition, the research results of the quantitative study highlighted the three types of chatbot affordances that facilitate chatbot use. The interviewees' quotations revealed several important technical features and functions that could further support our conceptualizations of the three chatbot affordances: enhancing customer service, supporting decision-making, and improving business agility. The quantitative study further confirmed these three types of chatbot affordances by validating the discriminant validity among them and proving the relationship between chatbot affordances and use. Online Supplemental Appendix 13 summarizes the meta-inferences, adapted

from Venkatesh et al. [77]. Our results demonstrate that the growth of AI might not lead to the elimination of jobs but might result in changes to the support structures available to augment human performance.

Theoretical Contributions and Implications

Enhancing the Understanding of Chatbots

Our study contributes to the chatbot literature by developing and empirically validating a context-specific model of the impacts of chatbots on employees' perceived work performance. Our model indicates that the IT artifact can be an appropriate theoretical lens to illuminate the mechanism through which chatbots generate the effects of IT [3, 51]. This finding indicates that the IT artifact needs to be considered when investigating emerging IT tools such as chatbots. The results highlighted the key aspects of the IT artifact, namely, chatbot affordances and support, which have not been examined to clarify the effects of IT on work performance in the existing literature, especially regarding chatbots.

Accordingly, this study provides a contextualized understanding of work performance by developing and validating the context-specific social support and technology affordances of chatbots from the IT artifact perspective. Following guidelines of a single-context theory, our study extends the literature concerning IT-enhanced work performance by developing a contextualized model of chatbots and examining the context-specific variables through which perceived work performance can be enhanced by the use of chatbots [32]. We grounded our study in the concepts of technology affordances and social support from the IT artifact perspective to help establish the relationship between chatbots and perceived work performance. The qualitative study helped contextualize the general theory, and on this basis we conceptualized the key variable, chatbot support, into two context-specific types: informational and emotional. It also helped us generate initial ideas about three higher-level technology affordances—enhancing customer service, supporting decision-making, and improving business agility—which allowed us to conceptualize and develop three context-specific types of chatbot affordance. The survey results confirmed these variables and relationships. This study allowed us to better understand the impact of chatbots on work performance, which we elaborate in the following sections.

Furthermore, by showing chatbots can indeed enhance employee performance, our study contributes to the literature by providing important insights into how chatbots can augment employees' tasks that would be risky to replace because of potentially negative outcomes. Researchers have argued that replacing employees with AI could generate important impacts on the workplace, and they suggest that future studies should explore the consequences of deploying AI in organizations, such as how employees feel about AI taking over tasks they previously performed [23]. Our study takes a different perspective and assesses how employees perceive chatbots that help them complete tasks that would be risky to assign solely to chatbots. Our study can add to the ongoing debate about the future of AI in the workforce by assessing AI-enabled chatbots from the augmentation angle [20] and by proposing mechanisms that support effective worker augmentation through AI-enabled tools. Specifically, employees can explore how to take advantage of chatbots to support their tasks and augment their performance. Future studies can assess how organizations can

redesign jobs and allow employees to focus more on tasks that would be too risky to assign to chatbots alone.

Developing Measures for Chatbot Affordances

Our contextualized model and results further highlight the important role of chatbot affordances in driving the effectiveness of chatbots. Our study focused on three second-order chatbot affordances. Our results showed that, although these three chatbot affordances are all positively related to chatbot use, the affordance for enhancing customer service had the strongest effect. Furthermore, one of the robustness checks (Figure L1 in Online Supplemental Appendix 12) showed that the affordances for enhancing customer service and supporting decision-making can increase both routine and innovative use, whereas the affordance for improving business agility is only positively related to innovative use. These are valuable findings that can provide an enhanced understanding of the motivation for implementing IT within organizations, given the different types of use.

New measures assessing the three chatbot affordances from the employee perspective could pave the way for future researchers to understand emerging IS phenomena, bringing technical concepts from other disciplines (e.g., computer science) via sociotechnical concepts (e.g., technology affordances). This method is important to fundamentally understand end-users' (e.g., employees') psychological perceptions and behaviors, which are critical to business researchers. Furthermore, to develop new items for chatbot affordances, we conducted a qualitative study via in-depth interviews with employees. This study offers practical insights to help researchers develop new items; in contrast, some prior studies have focused solely on the literature to develop measurement items (e.g., [45]).

Therefore, discussing chatbot affordances in terms of social and behavioral research can provide insight into how IS researchers can bring new perspectives or concepts from other disciplines to create new measures. This insight can also change the current thinking that emphasizes the study of chatbots from a technical viewpoint and lacks essential feedback from end-users (e.g., employees). Our study contributes to the literature by proposing relevant second-order technology affordances and assessing their effects on technology use in the context of chatbots. The proposed high-level affordances can serve as a bridge between specific technology features (e.g., low-level affordances) and business practices [13]. Future studies can also explore how different types of affordances impact employee performance in various tasks. It is possible that certain types of affordances are more important in one task but less important in another.

In addition, the three chatbot affordances could enhance our understanding of how to integrate chatbots into business processes such as customer services. For example, the affordance for supporting decision-making explains how individuals might use chatbots to improve their decisions during customer services. Although the technical architecture of specific chatbots might differ and evolve, the interactions between employees and chatbots provide a fundamental understanding of how chatbots might be incorporated into business processes. In this sense, chatbot affordances move beyond the technical capabilities and provide important insights into the future design and implementation of chatbots.

Extending Social Support Theory to Chatbots

Our research results show how informational and emotional support enhance perceived work performance. We conceptualized these two variables based on social support and

operationalized them to be centered more on IT within the context of chatbots, in response to the interviewees' comments in our qualitative study. This is different from the prior literature, which focused on measuring social support based on social interactions between individual users (e.g., [42, 82]) and explaining how social support can be facilitated or observed in social interactions or user-generated content. Our conceptualization of chatbot support has changed our fundamental understanding of social support and the mechanism through which it can be derived. Chatbot support provides important insights into how AI, as an emerging technology, can provide employees with efficient and effective support like that provided by humans.

In addition, recent studies have also started to pay attention to emotional support from chatbots [26, 53, 84]. Our study argues that emotional support has become an increasingly important factor that generates impacts on work performance. Our results (Figure 3 and Figure 12.2 in Online Supplemental Appendix 12) showed that chatbot-enabled emotional support plays a more important role than information support in improving perceived work performance. This indicates that emerging IT has changed the way technology generates impacts on employee performance. Our study complements prior studies concerning chatbots' anthropomorphism by assessing how chatbots can trigger emotion and provide emotional support [11, 17].

Emerging IT, such as AI and chatbots, can change the way technology impacts employees and organizations. In the past, IT had emphasized informational support as a way to affect employee performance. In today's emerging IT environment, emphasis should be placed on emotional support of employee performance. Using unique features, chatbots advance the traditional IT artifact to include informational and emotional support, emphasizing the latter. In this sense, our study can advance social support theory in relation to chatbots by going beyond the existing literature and validating new conceptualizations, supported by both the qualitative and quantitative data. Our study also indicates that social support can be facilitated in different ways based on the specific IT and level of use (e.g., individual and organizational). Accordingly, future studies should further examine the formulation of social support in different IT contexts, given that social support has played an increasingly important role in generating the effect of IT. Future studies could explore how chatbots' emotional support could help augment employees in different tasks.

Practical Implications

Our study also has several important practical implications. Overall, our results show that chatbots can provide opportunities to augment employees and their work performance. First, because both informational and emotional support are significantly related to perceived work performance, management might want to emphasize these two types of social support by implementing chatbots within their organization. For instance, management might take advantage of chatbots by providing employees with strong informational support by generating relevant information and knowledge, such as documents and reports.

Management might also educate and train employees to better incorporate such resources into their job duties, thus building their confidence to accomplish tasks. Further, one of the robustness checks (Figure 12.2 in Online Supplemental Appendix 12) showed that emotional support is more important for enhancing perceived work performance, especially workplace creativity. Therefore, organizations should focus on how

chatbots can provide emotional support when marketing employees use them to provide customer service. Chatbots can be implemented to provide reliable and customized service so employees feel comfortable integrating them. However, informational support should not be ignored because it still improves employees' workplace productivity.

Furthermore, our proposed high-level affordance can also provide guidelines for how to enhance informational and emotional support via chatbot use. The affordance for improving business agility has a strong effect. Therefore, when implementing chatbots, organizations need to focus on the features that improve marketing strategy. For example, a new analytic approach can be used to analyze data collected by chatbots to understand customer preferences and design promotion strategies. The affordances for enhancing customer service and supporting decision-making also have significant effects. Therefore, these two types of affordances should not be ignored. Marketing employees can analyze data collected by chatbots to help them create customer profiles and better serve customers. Enhanced decisions regarding offers and promotions could then be made to increase the efficiency and effectiveness of marketing campaigns.

Those findings offer useful insights not only for the implications of chatbots within organizations but also for design principles. Based on an enhanced understanding of chatbot affordances, vendors might improve their design features to create appropriate affordances and meet the needs of organizations.

Limitations and Future Research Directions

Our study has a few limitations. First, a survey company recruited the sample for the quantitative study. Although our participants' backgrounds were diverse, it is still possible our sample might not represent all marketing employees. Second, our subjects were marketing employees. Because chatbots in different departments can be used differently, it is possible that chatbots might have different effects on employee job performance across departments. Future studies are needed to assess how chatbots influence employee job performance in other contexts. Third, our study did not control for experience with chatbots. We did control for tenure, and employees with higher tenure might use chatbots for longer periods and better integrate chatbots into their tasks. However, we found no significant effect of tenure. Nevertheless, it is a limitation. Our study is also limited by not including chatbot training or technological self-efficacy as our control. Fourth, we measured work performance by the way that employees perceive their own performance. Future studies could focus on how employees' peers or supervisors perceive their work performance or use objective measures.

Our study focuses on chatbots' augmentation of employees' work performance, and we acknowledge that a complete reliance on chatbots for customer service can generate negative outcomes [5, 57]. Future studies could explore various negative outcomes of chatbots to further our understanding of chatbot deployment within organizations.

Future studies could also extend our study in different ways. First, because our study focuses on higher-level chatbot affordances, future studies could determine which specific lower-level chatbot affordances might actualize those high-level affordances.

Lin et al. [44] proposed three dimensions of chatbot affordance: automatability, availability, and personalization. It would be interesting to assess how such lower-level affordances

could generate the higher-level affordances proposed in our study. It would also be interesting to assess which other lower-level chatbot affordances could actualize the higher-level chatbot affordances of this study. Second, future studies could also explore other relationships depicted by Strong et al. [70], who argued that actions and outcomes have feedback effects on affordances. Longitudinal studies could be conducted in the future to understand how affordances arise over time as employees engage in different types of chatbot use.

Third, future studies could focus on other employees (e.g., human resource management) in organizations because they might use chatbots differently from marketing employees. Fourth, future studies could also validate our model in the context of other AI tools (e.g., virtual reality) and examine whether the results still hold. Although the second-order affordances identified in our study might not generalize to other contexts of AI, our study opens the door for studies to pay more attention to the augmentation effects of AI. Therefore, future research could be conducted to assess how different affordances (those in this study and new ones in additional contexts) augment individuals in various contexts. Lastly, future studies might explore different technical architectures of chatbots and other aspects such as channels and use cases.

Conclusion

Using a mixed-methods approach, we investigated how chatbots augment customer service employees' job performance from the IT artifact perspective. We used a qualitative study to develop context-specific variables including chatbot affordances (for enhancing customer service, for supporting decision-making, and for improving business agility) and chatbot support (informational and emotional support), which explains the mechanism through which chatbots improve work performance. We conducted a follow-up quantitative study, the results of which validated those contextualized variables and the research model. Our study contributes to the theoretical framework on this new chatbot phenomenon and provides helpful practical implications for organizations.

Notes

1. http://www.youtube.com/watch?v=1mP5e5-dujg
2. Our study focuses on AI-enabled chatbots. We provide a more detailed explanation below in the section "Chatbot: A Review From a Technological View."
3. Interested readers can refer to Adamopoulou and Moussiades [2] and Cahn [14] for further technical details.
4. http://www.gartner.com/reviews/market/enterprise-conversational-ai-platforms/vendor/amazon-web-services/product/amazon-lex/alternatives
5. https://aws.amazon.com/lex/faqs/?nc=sn&loc=6

Disclosure statement

No potential conflict of interest was reported by the author(s).

Notes on contributors

Xiaolin Lin is an Assistant Professor of Information Systems in the College of Business at California State University, Sacramento. He holds a PhD in Management Information Systems from

Washington State University. Dr. Lin's research focuses on the implications of information technology (IT) in e-commerce, human-AI interaction, cyber security, and gender differences in IT behavioral research. His work has been published in such journals as *Journal of Business Ethics, European Journal of Information Systems, Decision Sciences, Information & Management*, and *Journal of the Association for Information Science and Technology*, among others. He serves as a senior editor at *Information Technology & People*.

Xuequn (Alex) Wang is a Senior Lecturer in Edith Cowan University, Australia. He received his PhD in Information Systems from Washington State University. Dr. Wang's research interests include social media, privacy, e-commerce, and human-computer interaction. His research has appeared or forthcoming in *MIS Quarterly, Journal of Operations Management, Journal of the Association for Information Systems, European Journal of Information Systems*, and *Information Systems Journal*, among others.

Bin Shao is a Professor of Decision Management & Terry Professor of Business in the Paul and Virginia Engler College of Business, West Texas A&M University. She earned her PhD from University of Illinois at Urbana-Champaign. Dr. Shao's current research interests are mainly in the interdisciplinary areas of Information Systems, Decision Management, and Marketing. Her works have been published in International *Advances in Economic Research, International Journal of Applied Management Science, Journal of Supply Chain and Operations Management, Journal of Management Information and Decision Sciences*, and many others.

Joseph Taylor an Associate Professor of Information Systems in the College of Business at California State University, Sacramento. Prior to joining academia, he worked extensively in industry focusing on IT strategy, innovation and governance. His academic research interests focus on emerging tools and platforms as a means to enable technology to create business value and has appeared in such outlets as *Information Systems Journal, Journal of Strategic Information Systems*, and *Management Information Systems Quarterly Executive*.

ORCID

Xiaolin Lin http://orcid.org/0000-0001-8556-3889
Xuequn Wang http://orcid.org/0000-0002-1557-8265

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